



Smriti Machine Learning & Bhashini AI Architecture

Neurocognitive Rationale, Adaptive CST Calibration, Digital India Bhashini Speech Pipeline & Render Cloud Gateway

ML CLOUD MICROSERVICE
FastAPI on Render Cloud

SPEECH SYNTHESIS
Digital India Bhashini TTS

COGNITIVE DOMAINS
5 Clinical CST Domains

RESILIENCE PROTOCOL
6s Timeout + Heuristic Fallback

1. The Core Scientific Rationale: Why Machine Learning in Dementia?

Dementia is not a static condition; it is a progressive neurodegenerative syndrome with severe intra-day cognitive fluctuations (e.g., *Sundowning*, fatigue, and episodic lucidity). Static digital games fail dementia patients due to two catastrophic extremes:

THE FAILURE OF STATIC GAMES

The "Cognitive Wall" & Anxiety

When difficulty is static and exceeds the patient's momentary cognitive reserve, patients experience severe frustration, agitation, and task abandonment, accelerating cognitive withdrawal.

THE CLINICAL ML SOLUTION

Adaptive Flow & Neuroplasticity

Smriti uses ML to continuously calibrate the cognitive envelope (Yerkes-Dodson Law of Arousal). Tasks remain in the optimal "*Zone of Proximal Development*" to maximize synaptic stimulation without distress.

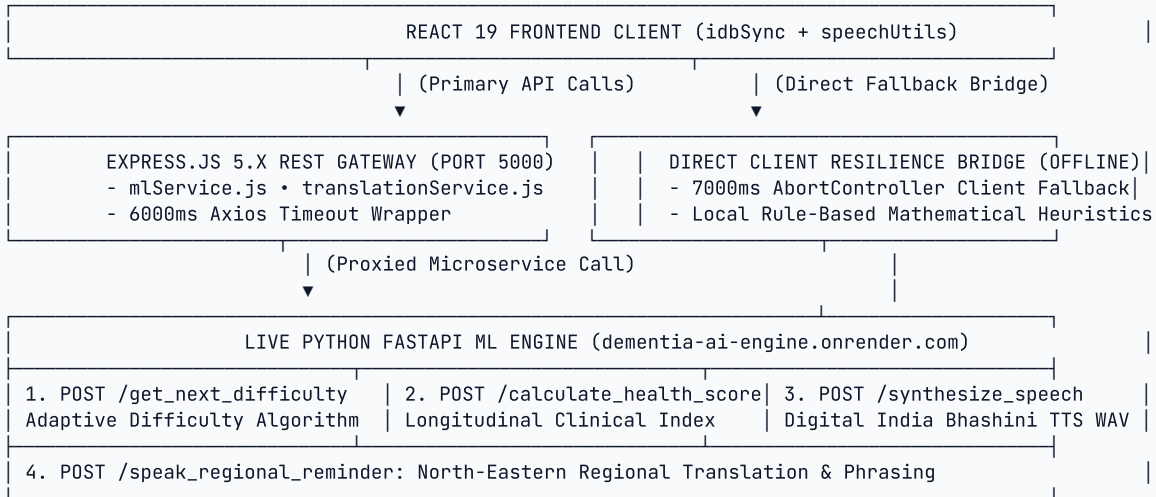
Key Clinical Objectives Solved by Smriti ML

- Real-Time Reaction Velocity Tracking (Δt):** Millisecond latency changes in visual/auditory recall serve as a sensitive digital biomarker of microvascular ischemia weeks before gross HMSE/MMSE score drops.
- Longitudinal Cognitive Index (S_{cog} in $[0, 100]$):** Replaces subjective caregiver recollection with standardized, objective telemetry aggregated across 7-day CST playthroughs.
- Zero-Barrier Regional Voice Inclusivity:** Digital India Bhashini TTS synthesizes native Assamese, Bodo, and Bengali speech in the cloud, removing reliance on missing local smartphone voice packs.

2. Render Cloud Microservice Architecture (dementia-ai-engine.onrender.com)

The ML microservice is deployed as an independent, high-throughput **Python FastAPI** service hosted on Render Cloud, decoupled from the Node.js business gateway to ensure zero CPU contention during mathematical evaluations.

RENDER CLOUD ML & BHASHINI GATEWAY TOPOLOGY



3. Digital India Bhashini Speech & Translation Pipeline

The biggest barrier to digital healthcare in North-East India is the **linguistic deficit**: commercial OS platforms (Android/iOS) do not bundle native Text-to-Speech (TTS) voice models for Assamese, Bodo, Manipuri, or tribal dialects. Smriti solves this by integrating Government of India **Bhashini (ULCA)** speech pipelines directly in our Render ML engine.

CLOUD SYNTHESIS

Base64 WAV Audio Streaming

The microservice converts text into high-fidelity 24kHz Base64-encoded WAV/MP3 audio buffers, returned in JSON payload for instantaneous HTML5 Audio playback in browser.

DUAL-ENGINE RESILIENCE

Zero-Crash Fallback Protocol

If the user is offline or the Bhashini API times out (>6s), the client smoothly fails over to local Web Speech API (English/Hindi) or displays the localized Assamese notice banner.

Bhashini API Request & Response Specification

```
// 1. INBOUND REQUEST: POST https://dementia-ai-engine.onrender.com/synthesize_speech
{
  "text_to_speak": "নমস্কাৰ ৰমেশ ডাঙৰীয়া, আপোনাৰ ৰাতিপুৱাৰ ঔষধ খোৱাৰ সময় হ'ল।",
  "target_language": "as" // Supported: 'as' (Assamese), 'hi' (Hindi), 'bn' (Bengali), 'brx' (Bodo)
}

// 2. OUTBOUND RESPONSE (JSON):
{
  "original_text": "নমস্কাৰ ৰমেশ ডাঙৰীয়া, আপোনাৰ ৰাতিপুৱাৰ ঔষধ খোৱাৰ সময় হ'ল।",
  "target_language": "as",
  "spoken_text": "নমস্কাৰ ৰমেশ ডাঙৰীয়া, আপোনাৰ ৰাতিপুৱাৰ ঔষধ খোৱাৰ সময় হ'ল।",
  "audio_base64": "UkLGRrZbBAXQVZFZm10IBIAAAADAAEAILYAAIhYQAECACAAAABkYXRh...", // WAV 24kHz Base64
  "engine": "bhashini_tts_engine",
  "source": "live_bhashini_api"
}

// 3. FRONTEND CONSUMPTION:
const audio = new Audio("data:audio/wav;base64," + response.audio_base64);
audio.play(); // Crystal clear native Assamese spoken playback on any device!
```

4. Core Mathematical Models & Algorithmic Formulations

Algorithm A: Dynamic CST Adaptive Difficulty (POST /get_next_difficulty)

Calculates the next starting difficulty tier D_{next} in $\{1, 2, 3, 4, 5\}$ at the end of each game level based on reaction latency Δt and cumulative mistake count M :

Performance Metric	Condition / Threshold	Difficulty Scaling	Clinical Rationale
High Precision & Speed	$M = 0$ AND $\Delta t < 3.5s$	$D_{next} = \min(5, D_{current} + 1)$	Elevates cognitive challenge to stimulate synaptic reserve.
Cognitive Fatigue / Stress	$M \geq 3$ OR $\Delta t > 7.0s$	$D_{next} = \max(1, D_{current} - 1)$	De-escalates complexity to avoid anxiety and cognitive overload.
Steady Performance	$1 \leq M < 3$ AND $3.5s \leq \Delta t \leq 7.0s$	$D_{next} = D_{current}$	Reinforces neuroplastic stabilization in current tier.

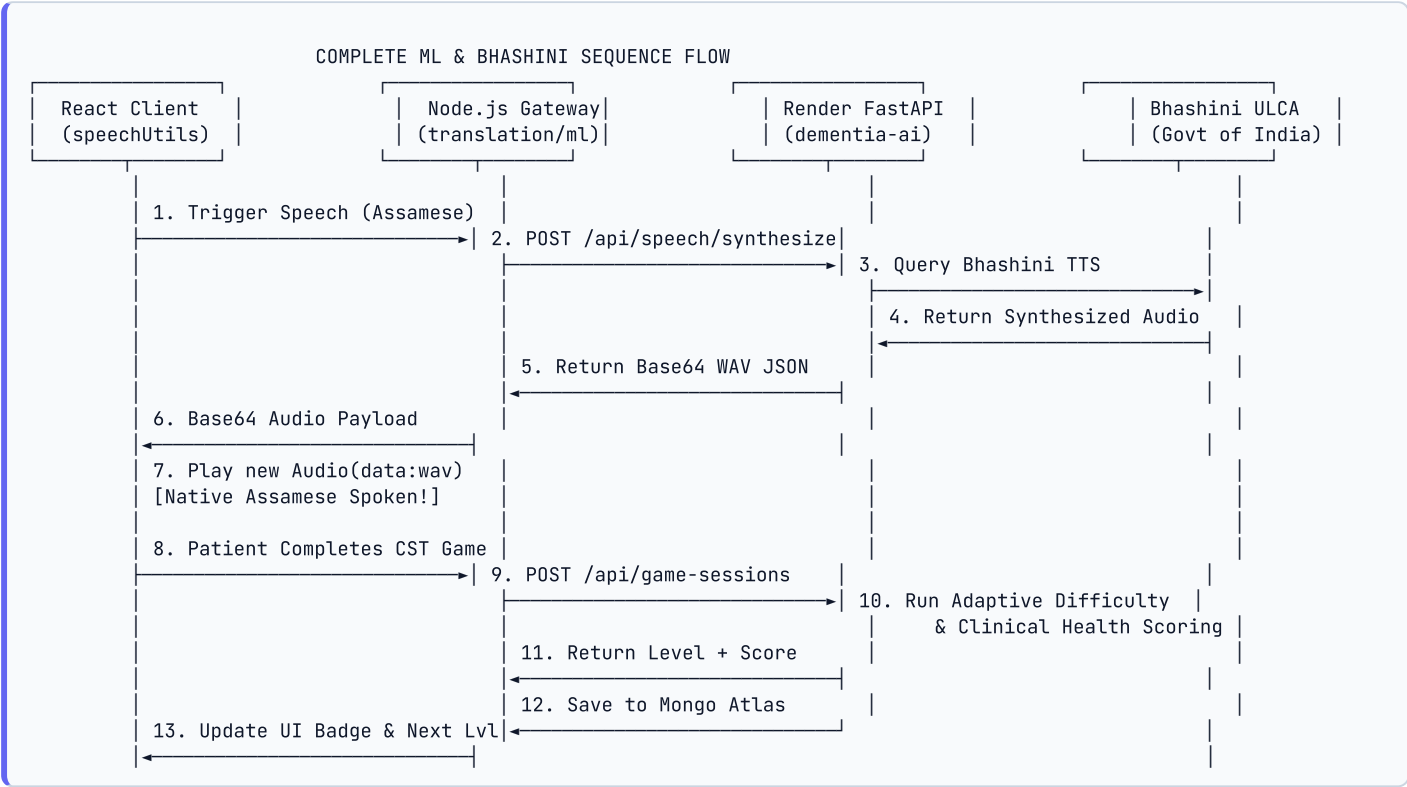
Algorithm B: Longitudinal Cognitive Health Scoring (S_{cog} in $[0, 100]$)

Synthesizes a 7-day moving window of cognitive sessions (N_{games}), average reaction time (Δt_{avg}), and mistake count (M_{total}):

```
Cognitive Health Score Formulation:
S_raw = 80 + min(15, N_games * 3) - min(30, M_total * 2.5) - min(15, max(0, avg_reaction - 3.0) * 3)
S_cog = clamp(S_raw, 40, 100)

Clinical Status Classification:
• S_cog ≥ 85 : "Stable" (Optimal CST Engagement)
• 70 ≤ S_cog < 85 : "Mild Decline - Monitor" (Requires Caregiver Attention)
• S_cog < 70 : "Significant Decline - Alert Doctor" (Triggers High-Priority Red Flag Alert)
```

5. End-to-End ML Telemetry & Audio Synthesis Sequence Flow



6. 3-Layer Fault Tolerance & Offline Fallback Matrix

Failure Mode	Primary Detector	Immediate Automated Recovery	User Experience Impact
Render ML Microservice Cold-Start / Latency	Node.js 6000ms Axios Timeout	Executes in-memory heuristic math functions in mlService.js.	Instant response (<5ms); difficulty and score calculated deterministically.
Bhashini Speech API Down / Network Drop	Frontend AbortController Timeout (7000ms)	Falls back to browser Web Speech API (English/Hindi) or Assamese banner.	Patient receives spoken voice in fallback language; no app lockup.
Complete Offline Outage (No Internet in Rural NER)	Service Worker + navigator.onLine === false	IndexedDB (idb) stores telemetry; local heuristics compute difficulty.	100% full gameplay offline; telemetry auto-syncs when online.